XM Satellite Weather System – General data

1. General

Provision for the XM Satellite Weather System to be installed as a customer standard option order 6211.

The XM Satellite Weather System is a streaming weather data receiver.

2. Description

A. XM Satellite Weather Receiver (XMD157–01)

The XM Weather receiver is mounted on the rear avionics rack. The XM Satellite Weather system sends received weather data to the Modular Avionics Unit (MAU) Advanced Graphics Unit (AGU), which is then displayed on the Multi-Function Display (MFD).

Basic functionality received by the XM Satellite and displayed on the MFD includes:

- Next Generation Radar (NEXRAD)
- Temporary Flight Restrictions (TFR)
- Airman's Meteorological Advisories (AIRMET)
- Significant Meteorological Information (SIGMET)
- Storm tops
- Lightning
- Winds
- Satellite cloud heights
- Turbulence
- METAR
- Continental United States (CONUS) radar coverage mask (part of NEXRAD)
- Echo tops

The XM Satellite Weather antenna is installed on the top of the fuselage at FS315.

28 Volt DC power is provided from the left bus through a 5 amp SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The XM Weather receiver interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

3. Adjustment/Test

A. Function Test XM Satellite Weather Receiver (XMD157–01)

1) General

These procedures are for the function test of the XM Satellite Weather System.

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel is closed.

d) Select XM Satellite weather to be displayed on the lateral map display.

e) Verify XM WX annunciator flashes in reverse video until the initial streaming data is received. The XM WX annunciator turns amber if no streaming data has been received for 10 minutes.

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – XM Satellite Antenna (XMRANT–01)

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the ceiling compartment forward of station 321.6. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

3) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the XM antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

4) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

5) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

6) Replace ceiling compartment. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal/Installation of XM Satellite Weather Receiver (XMD157–01)

1) General

These procedures are for the removal and installation of the XM Satellite Weather receiver, if installed.

2) Removal

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the aft avionic shelf through the rear baggage compartment.

d) Disconnect the electrical and coaxial cables at the front of the XM Satellite weather receiver.

e) Remove the four screws that secure XM Weather receiver to the avionics shelf.

f) Remove unit from aircraft.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Gain access to the aft avionic shelf through the rear baggage compartment.

d) Align the configuration module mounting holes to the avionics shelf.

e) Reconnect connectors to the XM Satellite weather receiver. Ensure connections are secure.

f) Remove safety clip and close SATELLITE WX (K4) circuit breaker on the avionics circuit breaker panel.

g) Test the XM Satellite Weather receiver as per Para 3.A.2).

B. Removal/Installation of XM Satellite Weather Antenna (XMRANT–01)

1) General

These procedures are for the removal and installation of the XM SATELLITE antenna, if installed.

2) Removal

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Support the antenna and remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

g) Remove antenna from aircraft.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with four mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.B.

k) Remove the safety clip and replace SATELLITE WX (K4) (Right Bus) circuit breaker on the avionics circuit breaker panel.

l) Do a function test on the XM Satellite receiver as per Para 3.A.2).

5. Troubleshooting

A. XM Satellite Weather (XMD157–01)

1) If the XM Satellite weather is deemed inoperative check the following:

a) Ensure SATELLITE WX (K4) circuit breaker is closed.

b) Visually inspect XM Satellite Weather connector for damaged wires or contacts.

c) Visually inspect XM Satellite Weather antenna for corrosion and contaminants.

d) Ensure that the XM Satellite Weather connector is properly mated and secure.

e) Inspect antenna external surfaces for damage.

f) Check that the antenna is properly mounted and screws are tight.

g) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

h) If repair or service is required, send to appropriate rated avionics repair facility.